

MAT236 - Homework 5 (SENT ON 04.05.2026)

Homework: Today (04/05/2026) we argued on the statement why $\sin z = 0 \implies z = k\pi, k \in \mathbb{Z}$ but $\arcsin z = 0, z = 2k\pi, k \in \mathbb{Z}$. The issue is addressed below.

1 The General Solution of $\sin(z) = 0$

The equation $\sin(z) = 0$ is satisfied for all values of z where the sine function vanishes. In the complex plane, this occurs only on the real axis at integer multiples of π :

$$z = k\pi, \quad k \in \mathbb{Z}$$

2 Logarithmic Definition of $\arcsin(z)$

The inverse sine function is defined using the complex logarithm as:

$$\arcsin(z) = -i \ln \left(iz + \sqrt{1 - z^2} \right)$$

Evaluating this at $z = 0$ yields:

$$\arcsin(0) = -i \ln \left(i(0) + \sqrt{1 - 0^2} \right) = -i \ln(\pm 1)$$

3 Resolve by Cases**3.1 Case A: The Positive Side**

If we take $\sqrt{1} = 1$, then $\arcsin(0) = -i \ln(1)$. Since the general complex logarithm is $\ln(1) = i(2k\pi)$, we have:

$$-i(i \cdot 2k\pi) = 2k\pi$$

This yields only the **even** multiples of π .

3.2 Case B: The Negative Side

If we take $\sqrt{1} = -1$, then $\arcsin(0) = -i \ln(-1)$. Since $\ln(-1) = i(\pi + 2k\pi)$, we have:

$$-i(i(\pi + 2k\pi)) = (2k + 1)\pi$$

This yields only the **odd** multiples of π .

4 Conclusion

$$z = \{2k\pi\} \cup \{(2k + 1)\pi\} = k\pi$$

In the class, we forgot to examine the negative side. The argument $\arcsin z = 0, z = 2k\pi, k \in \mathbb{Z}$ seems to be false. $\arcsin z$ is $\boxed{k\pi}$ periodic.